

HEPATITIS AND THE SKIN

There may be no visible skin changes in patients with severe liver disease, and conversely, dramatic cutaneous manifestations may develop in those with minimal hepatic dysfunction.

Cutaneous striae may appear in patients with chronic hepatitis, regardless of the underlying cause and even before glucocorticoid therapy (see Fig. 151-1). Pyoderma gangrenosum, urticaria, vasculitis, acne, scleroderma, and splinter nail hemorrhages can occur, depending on the specific etiology of the hepatitis.

Clinical Features of Acute Viral Hepatitis

The clinical features of infection with the different hepatotropic viruses that cause acute viral hepatitis are similar. In the pre-icteric phase, patients typically experience malaise, lethargy, nausea, and abdominal pain. A discrete, transient, maculopapular, urticarial, or petechial rash may develop, often accompanied by arthritis or arthralgia. These symptoms may worsen with the onset of jaundice, dark urine, and pale stools. Malaise and fatigue can persist for weeks or months after resolution of the acute illness.

Hepatitis A

Hepatitis A virus is transmitted via the fecal-oral route. Although relapses can occur, hepatitis A does not progress to chronic hepatitis.

Hepatitis B

Hepatitis B virus (HBV) is transmitted through percutaneous and permucosal routes. It is endemic in many regions, and healthcare professionals are at increased risk. A recombinant DNA vaccine is available, though it has been associated with cutaneous side effects.

Approximately 5% of HBV-infected individuals become chronic carriers, and up to 90% of infants born to infected mothers develop chronic infection.

There are four main cutaneous associations of hepatitis B:

1. **Serum sickness-like syndrome** – Occurs in about 10% of patients during the pre-icteric phase of acute HBV infection. Urticaria may be the dominant or only feature, but it is often accompanied by low-grade fever, arthralgia (especially in peripheral joints), proteinuria, and hematuria. Vasculitis may also be present.
2. **Cryoglobulinemia** – Both HBV and HCV are associated with cryoglobulinemia, though HCV is more commonly implicated. In HCV infection, mixed cryoglobulinemia is typically type III, and its presence may indicate active viral replication. Clinical features include recurrent purpura, acrocyanosis, arthralgia, lymphadenopathy, peripheral neuropathy, and hepatosplenomegaly. Renal involvement is common. Histopathology shows necrotizing vasculitis.
3. **Polyarteritis nodosa** – Can occur during acute HBV infection or develop years later. It is associated with circulating immune complexes containing hepatitis B surface antigen and immunoglobulin. Leukocytoclastic vasculitis may be present.
4. **Papular acrodermatitis of childhood (Gianotti-Crosti syndrome)** – Typically affects children and presents with monomorphic, flat-topped, erythematous papules on the face and limbs. The eruption is usually self-limiting and asymptomatic but may be associated with lymphadenopathy. Other infectious agents and vaccinations have also been linked to this condition.

Sweet syndrome has recently been reported in association with both acute and chronic HBV infection.

Hepatitis C

Hepatitis C virus (HCV) was historically the leading cause of posttransfusion hepatitis, though screening has significantly reduced new cases.

Cutaneous manifestations of acute and chronic HCV infection include:

- Leukocytoclastic vasculitis (often due to cryoglobulinemia)
- Cutaneous necrosis
- Pruritus
- Acquired angioedema
- Porphyria cutanea tarda (PCT)
- Erythema nodosum
- Urticaria
- Erythema multiforme
- Polyarteritis nodosa

Lichen planus is more commonly observed in HCV-infected patients, although study results vary regarding the exact incidence. An increased frequency of the HLA-DR6 allele has been noted in Italian patients with HCV-associated oral lichen planus, suggesting a role for host genetic factors in this association.